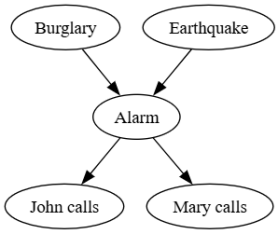
Exercise 9: Bayesian Networks

1. Plotting the network and its Laplacian (Task 1)



graphviz plot of the alarm network

The alarm network was drawn with graphviz. It has five nodes: Burglary (A), Earthquake (B), Alarm (C), John calls (D) and Mary calls (E). The edges are A→C, B→C, C→D and C→E.
The in-degree Laplacian is L = D_in − Adj, where Adj[i][j] = 1 if i→j. The in-degrees are A=0, B=0, C=2, D=1 and E=1.

$$L = \begin{bmatrix} 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 2 & -1 & -1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Rows and columns are in the order A–E.
Row C reads 2 on the diagonal because C has two parents, A and B.
Rows D and E read 1 for their single parent C.

2. Bayesian network construction and inference

The alarm network was built in sorobn with the conditional probability tables from alarm.png

Q1. Likelihood of a burglary if both John and Mary call
P(Burglary = True | John, Mary call) = **0.284**. Even with both neighbours calling, a burglary is still unlikely (72% False).

```
Burglary
False    0.715828
True     0.284172
Name: P(Burglary), dtype: float64
```

Q2. Chance that John and Mary call if an earthquake
P(John = True, Mary = True | Earthquake) = **0.183**. The other outcomes are John only 0.114, Mary only 0.027 and neither 0.676.

```
John calls  Mary calls
False       False         0.675854
            True          0.027085
True        False         0.113591
            True          0.183470
Name: P(John calls, Mary calls), dtype: float64
```

3. Practice network (Tasks 2 and 3)

Task 2. P(F = True | A, D true; B, C false)

The result is **0.2**.

```
F
False    0.8
True     0.2
Name: P(F), dtype: float64
```

Task 3. P(F = False | A, B false; C, D true)

The result is **0.8**.

```
F
False    0.8
True     0.2
Name: P(F), dtype: float64
```